

Dandy-Walker Malformation

Society for Maternal-Fetal Medicine (SMFM); Ana Monteagudo, MD

Introduction

Dandy-Walker malformation (DWM) is a complex malformation involving the posterior fossa and cerebellum. It is a rare condition with an estimated incidence of 1 in 10,000 to 30,000 births. Hydrocephaly, a common finding, is seen in approximately 80% of cases.¹ DWM is present in 4% to 12% of cases of hydrocephaly in infants.² Most cases of DWM are sporadic, although first-degree relatives have an increased risk of DWM compared with the general population.¹

Definition

The “classic triad” of DWM is defined by (1) complete or partial agenesis of the vermis; (2) enlargement of the posterior fossa with the upward displacement of the tentorium, transverse sinus, and torcular; and (3) cystic dilation of the fourth ventricle.^{3–5} The pathogenesis of DWM involves an intricate developmental anomaly of the cerebellar vermis in which the fourth ventricle fails to close, causing a persistent Blake’s pouch. The fourth ventricle enlarges, upwardly displacing the cerebellar vermis. As a result of this process, the development of the tentorium, straight sinus, and torcula is arrested, which causes the posterior fossa to enlarge.⁵

DWM is usually diagnosed on second-trimester ultrasound examination. The typical finding is an enlarged

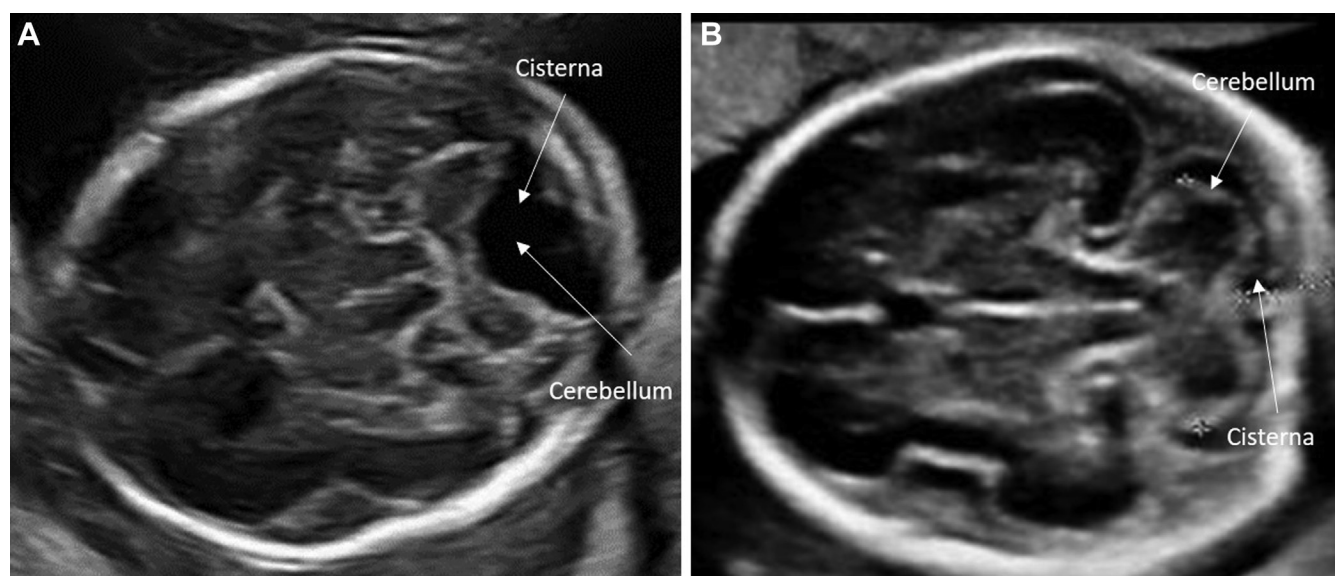
cisterna magna measuring >10 mm in the axial plane at the level of the transcerebellar diameter. Ventriculomegaly with lateral ventricles measuring ≥ 10 mm is a common finding. It has been hypothesized that the development of ventriculomegaly is secondary to dynamic changes in the cerebrospinal fluid related to the mass effect of the malformation.^{3,4}

During the first trimester of pregnancy, visualization of the posterior fossa is possible using both transabdominal and transvaginal ultrasonography.⁶ First-trimester diagnosis of DWM has been reported.^{6–8} Visualization of the cerebellum and cisterna magna with ultrasound is possible using sagittal and axial views of the fetal head from 11 3/7 weeks of gestation to 13 6/7 weeks of gestation. In a report of 4 fetuses in which DWM was confirmed at 18 to 22 weeks of gestation, the ratio of the brainstem (BS) diameter to the brainstem to occipital bone (BSOB) diameter (BS/BSOB) was measured in the sagittal view; this ratio was found to be significantly larger in cases of DWM when compared with normal controls.⁸

Ultrasound Findings

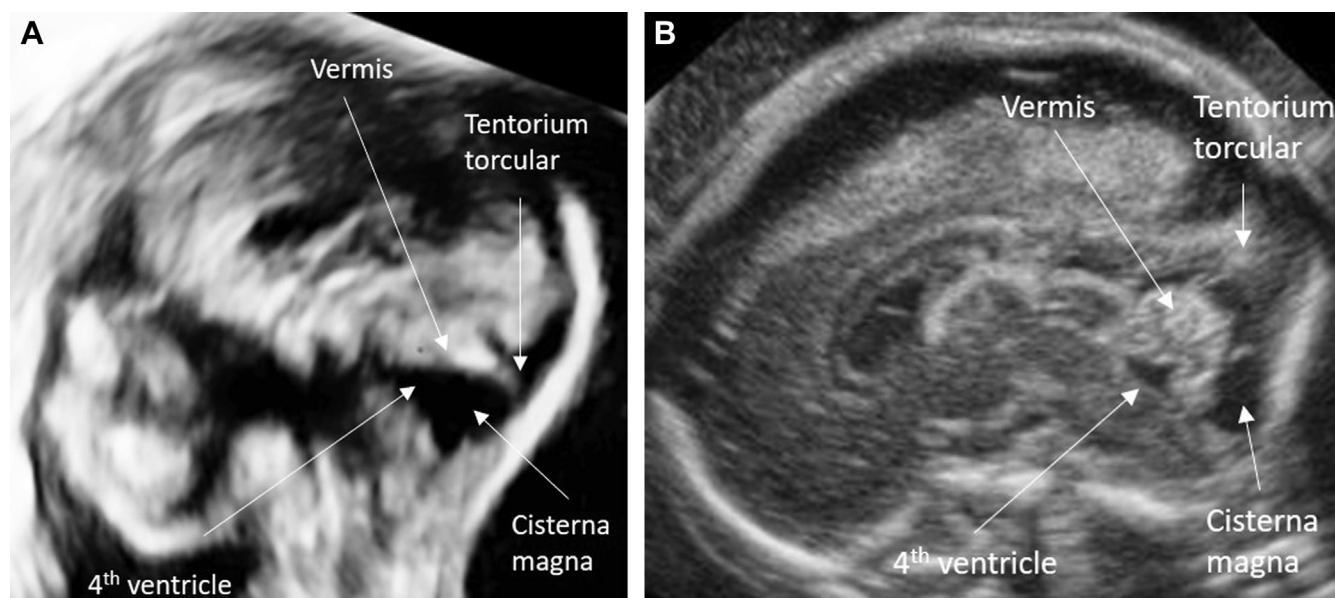
In the axial plane at the level of the transcerebellar plane, the cisterna magna is enlarged (>10 mm). The cerebellar

FIGURE 1
Dandy-Walker malformation vs a normal image



A, Axial section at the level of the transcerebellar plane. The cisterna magna is enlarged, and the cerebellar hemispheres are splayed apart. **B**, Normal image included for comparison.

SMFM. SMFM Anomalies Consult Series #3. Am J Obstet Gynecol 2020.

FIGURE 2**Median section obtained using transvaginal ultrasonography**

A, A fetus at 20 weeks of gestation with suspected DWM. The vermian is small and rotated. The fourth ventricle is enlarged and freely communicates with the cisterna magna. The torcular and tentorium are upwardly displaced. **B**, Normal image included for comparison.

SMFM. SMFM Anomalies Consult Series #3. Am J Obstet Gynecol 2020.

hemispheres are splayed apart, and the vermian is absent or hypoplastic (Figure 1). The lateral ventricles may show varying degrees of ventriculomegaly (defined as enlargement of the ventricles of ≥ 10 mm). In normal pregnancies of < 20 weeks of gestation, the connection between the fourth ventricle and the cisterna magna has not yet closed, and this connection may be seen on ultrasound. In these cases, a repeat ultrasound examination after 20 weeks of gestation is indicated. The presence of a connection between the fourth ventricle and the cisterna magna after 20 weeks of gestation is indicative of DWM.⁹

Visualization of the axial plane alone is insufficient to diagnose DWM.¹⁰ A median (sagittal) section of the posterior fossa can help differentiate DWM from other types of posterior fossa abnormalities, including Blake's pouch cyst, mega cisterna magna, and vermian hypoplasia. The median plane can be easily obtained using 2-dimensional or 3-dimensional transabdominal or transvaginal ultrasonography; if the fetus is in a cephalic presentation, transvaginal ultrasonography is the route of choice (Figure 2). In the median plane, the vermian can be easily identified because it is echogenic compared with the cerebellar hemispheres and sits on top of the brainstem. Once the vermian has been identified, the size of the vermian can be assessed either qualitatively or quantitatively. Nomograms of the vermian size are available; however, because most cases of an abnormal vermian involve the posterior portion, measuring the vertical diameter is essential to diagnosis.⁹ The appearance of the vermian is also important.¹¹ If the apex of

the fourth ventricle (fastigium) is normal and the 2 main fissures of the vermian are present, it can be surmised that the vermian is normal. A normal-appearing vermian that is too small suggests vermian hypoplasia. A small vermian that lacks a fastigium, fissure, or both suggests partial vermian agenesis.⁹ The position of the tentorium and torcular can be assessed; upward displacement of the tentorium and torcular results from an enlarged posterior fossa.

Some authors suggest measuring the brainstem-vermian (BV) and the brainstem-tentorium (BT) angle to categorize the upward rotation of the vermian, which may help differentiate among the most common pathologies involving the posterior fossa. The BV angle is obtained by drawing a line tangentially to the dorsal aspect of the brainstem and a second line tangentially to the ventral contour of the vermian. The BT angle is obtained by adding a third line tangentially to the tentorium. The BV and BT angles increase with increasing severity of the posterior fossa pathology; however, of the 2 measurements, the BV angle seems to be more useful. BV angles > 45 degrees strongly suggest DWM, whereas angles < 30 degrees are suggestive of Blake's pouch cyst.¹²

Associated Abnormalities

DWM is frequently associated with other central nervous system (CNS) and non-CNS anomalies and chromosomal and nonchromosomal syndromes. In postnatal studies, the frequency of associated malformations ranges from 50% to 70%.^{9,10}

The most common CNS anomalies associated with DWM include ventriculomegaly, agenesis of the corpus callosum, holoprosencephaly, and encephalocele.^{9,10} Other less common findings include migrational disorders and heterotopias. The most common non-CNS anomalies are congenital heart disease, polycystic kidneys, and facial clefts. Other less frequent anomalies include limb and abdominal wall abnormalities, diaphragmatic hernia, ambiguous genitalia, and fetal growth restriction.^{13,14}

Chromosomal abnormalities, such as trisomies 9, 13, 18, and 21; triploidy; and 6p and 3q22-q24 deletion, are associated with DWM; associated nonchromosomal syndromes include Meckel and Walker-Warburg syndromes. In a study of 187 cases of DWM, approximately one-third were reported to have a chromosomal abnormality or another genetic syndrome, the most frequent being PHACE syndrome (Posterior fossa, Hemangioma, Arterial abnormality, Cardiac malformations, Eye abnormalities); trisomy 18; and Joubert syndrome.¹ The prevalence of chromosomal abnormalities in fetuses with isolated DWM has been reported to be 16.3%, with deletions being the most common.¹⁰ Other reported associations are environmental exposures, such as alcohol and maternal diabetes, and infections, such as rubella and cytomegalovirus (CMV).^{9,15}

Differential Diagnosis

DWM must be distinguished from other posterior fossa cystic malformations, including Blake's pouch cyst, mega cisterna magna, vermian and cerebellar hypoplasia, and arachnoid cyst.³ Establishing the correct diagnosis is important because conditions such as Blake's pouch cyst and mega cisterna magna have a much more favorable prognosis than DWM and are less frequently associated with anomalies. Blake's pouch (cyst) is a normal embryologic structure in the development of the posterior fossa. One-third of such cases resolve during pregnancy, and a normal neurodevelopmental outcome has been reported in approximately 90%.¹⁶

The median plane of the brain plays a key role in the ultrasonographic evaluation and differential diagnosis of posterior fossa cystic malformations. The 3 most useful criteria for the diagnosis of Blake's pouch cyst are (1) normal anatomy and size of the vermis, (2) mild to moderate rotation of the vermis, and (3) normal size of the cisterna magna. Of note, Blake's pouch cyst resolves in more than 50% of the cases between 24 to 26 weeks of gestation as a result of delayed fenestration of the fourth ventricle.¹⁷ In mega cisterna magna or a cisterna magna measuring >10 mm, the vermis is normal in size and appearance, and there is no ventriculomegaly or hydrocephaly present. In vermian hypoplasia, the vermis is small but has a normal appearance. In cerebellar hypoplasia, the large cisterna magna is the result of the small cerebellum. With a posterior fossa arachnoid cyst, the cyst compresses the normal vermis anteriorly against the brainstem.¹⁸

Genetic Evaluation

Diagnostic testing (amniocentesis) with chromosomal microarray analysis (CMA) should be offered when DWM is detected. It is reasonable to initially perform karyotype analysis or fluorescence in situ hybridization, with reflex to CMA if these test results are normal. If there are additional anomalies, consanguinity, or a family history of a specific condition, gene panel testing or exome sequencing is sometimes useful because CMA does not detect single-gene (Mendelian) disorders. If exome sequencing is pursued, appropriate pretest and posttest genetic counseling by a provider experienced in the complexities of genomic sequencing is recommended. After appropriate counseling, cell-free DNA screening is an option for patients who decline diagnostic evaluation, although this will not detect the underlying genetic cause in most cases unless a common aneuploidy is suspected.

Pregnancy and Delivery Management

In cases of suspected DWM, a complete anatomic survey, detailed fetal neurosonogram, and fetal echocardiography are indicated as part of the initial evaluation. Fetal magnetic resonance imaging should be considered in cases in which a fetal neurosonogram is not feasible or to rule out other associated intracranial malformations. Amniocentesis should be offered for genetic studies and potentially for evaluation of fetal infections, such as rubella and CMV. Consultation with maternal-fetal medicine, neonatology, and pediatric neurology services is suggested. In general, pregnancy termination is an option that should be offered to patients when a major fetal anomaly is detected, including DWM.

If the patient continues the pregnancy, serial ultrasound examinations are suggested to monitor fetal growth and the degree of ventriculomegaly or hydrocephaly. Vaginal delivery is not contraindicated in cases of DWM, although cesarean delivery may be necessary in cases of macrocephaly resulting from hydrocephaly. If macrocephaly or severe hydrocephaly is present, cephalocentesis may be considered to achieve a vaginal birth in severe cases with a particularly poor prognosis.

Prognosis

Evidence regarding the long-term prognosis of fetuses or neonates with DWM is conflicting. One study reports that all children with DWM had some degree of cognitive impairment, whereas other studies report a more favorable outcome. Overall, it appears that up to one-third of survivors with DWM develop normally.¹⁹ Children with no associated CNS or extra-CNS anomalies and with a normal lobulated vermis have a more favorable neurodevelopmental outcome.

The prognosis depends on the presence or absence of associated malformations, the presence of chromosomal aneuploidy or genetic syndromes, and the degree of hydrocephaly. DWM can lead to several neurodevelopmental problems, which become apparent early in infancy. The

most common problems are macrocephaly, signs of increased intracranial pressure, spastic paraparesis, hypotonia, slow motor development, and intellectual disability.¹ It can also be associated with major psychopathology, such as schizophrenia, bipolar disorder, depression, and impulse control disorders.¹

A recent systematic review evaluating cases of isolated posterior fossa abnormalities on prenatal imaging included 13 infants with DWM, normal karyotype, and no other associated anomalies.⁴ The overall rate of abnormal neurodevelopment was 58%. Ventriculomegaly before or after birth occurred in 68% of the cases, with 63% requiring a ventriculoperitoneal shunt.⁴

Summary

DWM is an uncommon and heterogeneous malformation involving the cerebellar vermis. The first clue to the diagnosis is an enlarged cisterna magna. This condition is associated with CNS and non-CNS malformations and genetic conditions; therefore, genetic counseling and testing with karyotype analysis and CMA are suggested. A favorable long-term outcome of DWM has been reported in 30% to 42% of infants with no other abnormalities or genetic malformations.^{4,19}

REFERENCES

1. Stambolliu E, Ioakeim-Ioannidou M, Kontokostas K, Dakoutrou M, Kousoulis AA. The most common comorbidities in Dandy-Walker syndrome patients: a systematic review of case reports. *J Child Neurol* 2017;32:886–902.
2. Pilu G, Perolo A, Falco P, Visentin A, Gabrielli S, Bovicelli L. Ultrasound of the fetal central nervous system. *Curr Opin Obstet Gynecol* 2000;12:93–103.
3. Guibaud L, Larroque A, Ville D, et al. Prenatal diagnosis of 'isolated' Dandy-Walker malformation: imaging findings and prenatal counselling. *Prenat Diagn* 2012;32:185–93.
4. D'Antonio F, Khalil A, Garel C, et al. Systematic review and meta-analysis of isolated posterior fossa malformations on prenatal imaging (part 2): neurodevelopmental outcome. *Ultrasound Obstet Gynecol* 2016;48:28–37.
5. Shekdar K. Posterior fossa malformations. *Semin Ultrasound CT MR* 2011;32:228–41.
6. Egle D, Strobil I, Weiskopf-Schwendinger V, et al. Appearance of the fetal posterior fossa at 11 + 3 to 13 + 6 gestational weeks on transabdominal ultrasound examination. *Ultrasound Obstet Gynecol* 2011;38:620–4.
7. Garcia-Posada R, Eixarch E, Sanz M, Puerto B, Figueras F, Borrell A. Cisterna magna width at 11–13 weeks in the detection of posterior fossa anomalies. *Ultrasound Obstet Gynecol* 2013;41:515–20.
8. Lachmann R, Sinkovskaya E, Abuhamad A. Posterior brain in fetuses with Dandy-Walker malformation with complete agenesis of the cerebellar vermis at 11–13 weeks: a pilot study. *Prenat Diagn* 2012;32:765–9.
9. Pilu G, Malinge G, Buyukkurt S. Anomalies of the cerebellum. In: Timor-Tritsch IE, Monteagudo A, Pilu G, Malinge G, eds. *Ultrasonography of the prenatal brain*, 3rd ed. New York, NY: McGraw-Hill; 2012:283–302.
10. D'Antonio F, Khalil A, Garel C, et al. Systematic review and meta-analysis of isolated posterior fossa malformations on prenatal ultrasound imaging (part 1): nomenclature, diagnostic accuracy and associated anomalies. *Ultrasound Obstet Gynecol* 2016;47:690–7.
11. Malinge G, Ginath S, Lerman-Sagie T, Watemberg N, Lev D, Glezerman M. The fetal cerebellar vermis: normal development as shown by transvaginal ultrasound. *Prenat Diagn* 2001;21:687–92.
12. Volpe P, Contro E, De Musso F, et al. Brainstem-vermis and brainstem-tentorium angles allow accurate categorization of fetal upward rotation of cerebellar vermis. *Ultrasound Obstet Gynecol* 2012;39:632–5.
13. Harper T, Fordham LA, Wolfe HM. The fetal Dandy Walker complex: associated anomalies, perinatal outcome and postnatal imaging. *Fetal Diagn Ther* 2007;22:277–81.
14. Salihu HM, Kornosky JL, Druschel CM. Dandy-Walker syndrome, associated anomalies and survival through infancy: a population-based study. *Fetal Diagn Ther* 2008;24:155–60.
15. Imataka G, Yamanouchi H, Arisaka O. Dandy-Walker syndrome and chromosomal abnormalities. *Congenit Anom (Kyoto)* 2007;47:113–8.
16. Gandolfi Colleoni G, Contro E, Carletti A, et al. Prenatal diagnosis and outcome of fetal posterior fossa fluid collections. *Ultrasound Obstet Gynecol* 2012;39:625–31.
17. Paladini D, Quarantelli M, Pastore G, Sorrentino M, Sglavo G, Nappi C. Abnormal or delayed development of the posterior membranous area of the brain: anatomy, ultrasound diagnosis, natural history and outcome of Blake's pouch cyst in the fetus. *Ultrasound Obstet Gynecol* 2012;39:279–87.
18. Bornstein E, Monteagudo A, Timor-Tritsch IE. Intracranial cysts. In: Timor-Tritsch IE, Monteagudo A, Pilu G, Malinge G, eds. *Ultrasonography of the prenatal brain*, 3rd ed. New York, NY: McGraw-Hill; 2012:351–72.
19. Bolduc ME, Limperopoulos C. Neurodevelopmental outcomes in children with cerebellar malformations: a systematic review. *Dev Med Child Neurol* 2009;51:256–67.